

Detecting Multivariate Time Series Anomalies With Cascade Decomposition Consistency

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Abstract—Multivariate time series anomaly detection is crucial in sensitive domains such as cybersecurity and grid monitoring, significantly contributing to the reliability and safety of system operation. However, current methods suffer from inadequate utilization of decomposed time series, insufficient mining of contextual dependencies within the time series, and limited robustness against anomalies during training. To address these limitations, we propose the consistency-enhanced normalizing flow (ConFlow) model, which utilizes the consistency of decomposed time series and contextual temporal embedding to enhance the discriminative ability of the flow model. First, to refine the extraction of time series components, we propose a cascade decomposition and mixing module that iteratively decouples the time series. Second, these components are mapped to Gaussian distributions through the context-aware normalizing flow, incorporating both inter- and intra-series information into the density estimation. Third, the density consistency among decomposed time series is measured to reweight the estimation, while highly inconsistent series are viewed as anomalies and masked during training to improve model robustness. Finally, anomalies are detected using reweight density estimation. Experiments on five widely used datasets in the time series anomaly detection field demonstrate the superiority of our method over state-of-the-art (SOTA) approaches.

Index Terms—Consistency measurement, density estimation, normalizing flow, time series anomaly detection, time series decomposition.

I. INTRODUCTION

TIME series exist in various forms in different domains, such as electroencephalogram (EEG) recordings [1], industrial sensor outputs [2], and flight trajectories [3]. The ability to detect anomalies in time series data is paramount for ensuring operational efficiency and reliability [4]. For example, in motor fault diagnosis [5], time series anomaly detection facilitates the early identification of irregular motor behaviors, preventing unexpected downtime and reducing maintenance

costs through predictive maintenance. Similarly, in healthcare, the early detection of anomalies in heart rate variability, blood glucose levels, and respiratory patterns can be lifesaving for patients, facilitating timely medical interventions [6].

The early methods of anomaly detection relied on statistical techniques such as extreme value theory [9] and statistical tests [4], [10]. However, as data has become high-dimensional and non-stationary, these statistical methods have become less effective due to their strict assumptions about data distribution. To overcome these drawbacks, prediction-based [11], [12] and reconstruction-based methods [13] that are empowered by deep learning methods have emerged. These techniques rely on clean data to learn normal representations and identify data with significant prediction or reconstruction errors as anomalies. However, labeled data is costly, and clean datasets are often difficult to obtain, which limits the adaptability of these methods. In contrast, density-based methods are less dependent on labels and quality of training data, identifying data distributed in the sparse regions as anomalies through density estimation. Nevertheless, classical techniques in this category, such as kernel density estimation (KDE) [10] and Gaussian mixture models (GMMs) [14], face challenges in effectively managing high-dimensional time series.

In recent years, density-based approaches have turned to normalizing flow models, a type of deep generative model that enables flexible density estimation through an invertible transformation [16], [17]. Normalizing flows were first introduced for time series anomaly detection by GANF [18], which augments the flow model with a directed acyclic graph to better capture temporal dependencies, thus improving the accuracy of the estimation. However, GANF estimated the density for all series using a unified distribution, ignoring the specific characteristics of individual series. To address this limitation, MTGFLOW [19] was proposed, mapping different time series into distinct Gaussian distributions. However, MTGFLOW overlooked intra-series temporal information, limiting the estimation accuracy of the normalizing flow. The above methods still share some limitations that need improvement.

- 1) *Insufficient utilization of decomposed time series*: Time series can be decomposed into low-frequency trend and high-frequency seasonal components (as shown in Fig. 1) [20], offering detailed and diverse views for temporal analysis [21]. However, effectively utilizing decomposed components to enhance anomaly representation in density-based methods remains under-explored.
- 2) *Insufficient mining of contextual dependencies in time series*: Anomalies typically exhibit different correlations

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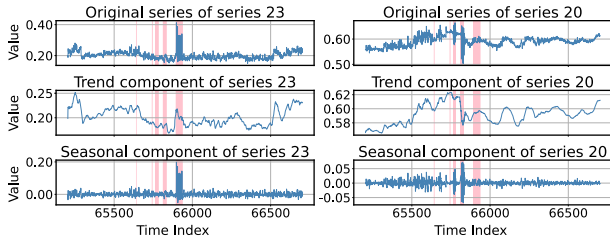


Fig. 1. Original time series as well as the trend and seasonal components extracted from it through seasonality-trend decomposition [7]. Both of the original series are selected from the PSM dataset [8].

with their adjacent time points [22]. Current anomaly detection methods primarily focused on modeling temporal correlation between different series but overlooked the contextual information within individual time series, degrading the accuracy of density estimation and anomaly detection.

- 3) *Limited robustness against anomalies during model training*: Normalizing flows do not depend heavily on a clean training set. However, a training set with many abnormal time series may cause models to misinterpret frequent anomalous patterns as normal. Therefore, the strategy should be designed to enhance model tolerance against anomalies.

To address the above challenge, we propose consistency-enhanced normalizing flow (ConFlow). Compared with previous works that based on the normalizing flow model [18], [19], we further utilize time series decomposition and find that abnormal decomposed time series often show poor consistency in density estimation (details elaborated in Section III). We utilize this generalizable characteristic of anomalies and design a consistency-mask loss reweight module to enhance model robustness and discriminative ability. Additionally, we encode contextual relevance within individual time series using self-attention, which captures subtle variations in time series so as to help anomaly detection. Overall, our model contains three key innovative components. First, a cascade decomposition and mixing module to iteratively refine the discriminative decomposed components. Second, context-aware normalizing flow that embeds both inter- and intra-series correlation for accurate density estimation. Third, a consistency-mask loss reweight module that utilizes the consistency of decomposed components to enhance model robustness and discriminative ability.

Our main contributions can be summarized as follows.

- 1) We propose an enhanced normalizing flow model that utilizes cascade decomposition consistency and contextual dependencies within time series to improve detection accuracy and robustness.
- 2) We incorporate a self-attention mechanism to capture contextual correlations and design a loss reweight and mask module that utilizes the density inconsistency of anomalies to enhance model performance.
- 3) We validated the effectiveness of our approach on five benchmark datasets, demonstrating superior performance compared to eight state-of-the-art (SOTA) time series anomaly detection methods.

This article is organized as follows. Section II reviews the related work on time series decomposition and anomaly detection. Section III details our findings on the density inconsistency of anomalous time series and provides some background information on the normalizing flow models. Section IV presents a detailed description of the proposed model, followed by extensive experiments that demonstrate the performance of the proposed model in Section V. Finally, Section VI summarizes this article and presents some limitations of our work.

II. RELATED WORKS

Considering that time series decomposition serves as the foundation of ConFlow, we survey the approaches of time series decomposition before reviewing the time series anomaly detection methods.

A. Time Series Decomposition

Time series typically consist of multiscale periods and diverse underlying patterns [20]. Decomposition techniques can help to separate these patterns [23], thereby improving data utility and feature representation quality. In recent years, decomposition has become integral to various time series tasks such as forecasting, classification, and anomaly detection, contributing to several SOTA methods like TimeMixer [24] and FEDformer [25].

Commonly used time series decomposition methods include discrete wavelet transformation [26] and seasonality-trend decomposition [27], both of which divide time series into high-frequency seasonal components and low-frequency trend components [28]. Wu proposed a seasonality-trend decomposition block [7] using moving average pooling to separate trend and seasonal components. This method is simple to implement and integrate with deep learning models, offering a fast and scalable solution for time series decomposition.

Recently, the time series decomposition module has been incorporated into several prediction-based and reconstruction-based time series anomaly detection methods, such as MEGA [20] and TimesNet [29], achieving improved performance by combining anomaly scores across different decomposed components. However, effectively utilizing decomposed time series in density-based anomaly detection methods remains a challenge. Decomposition often leads to varying density estimations and inconsistent anomaly scores, as shown in Fig. 2. Simply combining derived anomaly scores from the original series and its decomposed components can produce confounded results and achieve poor anomaly discriminative ability [as shown in Fig. 2(d)]. Thus, exploring a generalizable anomaly representation of decomposed series is essential to enhance the discriminative ability of density-based methods [30].

B. Time Series Anomaly Detection

Generally, time series anomaly detection models can be classified into supervised and unsupervised ones. Supervised models rely on high-cost human labels, thus making

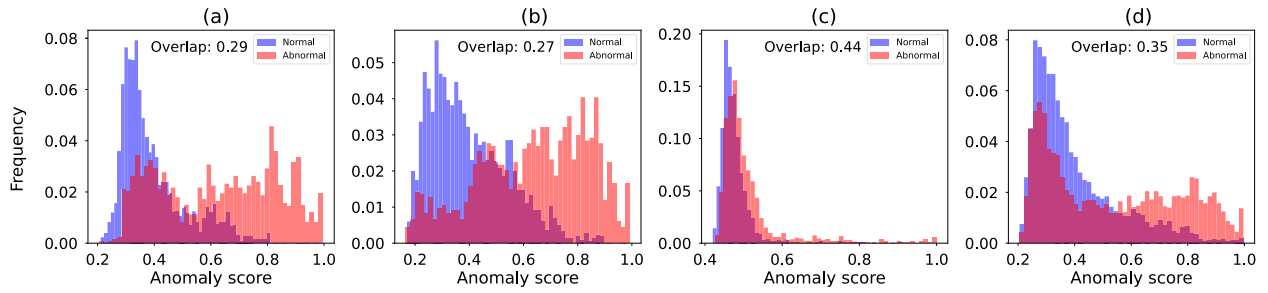


Fig. 2. Normalized anomaly scores on PSM test set [8] achieved by the MAF [15] for (a) original, (b) trend, and (c) seasonal time series. (d) Score accumulation depicts the normalized score sum of (a)–(c). The overlap (magenta) between the normal (indigo) and abnormal (salmon) histograms indicates the discriminative ability of the model.

unsupervised methods preferable in most cases. Specifically, unsupervised methods can be broadly classified into three types.

- 1) Prediction or reconstruction-based methods, which aim to learn representations of normal sequences and identify anomalies based on prediction [31] or reconstruction errors [32], [33]. Recent advanced architectures like Transformer enhance these methods with improved global and local perception ability on time series, thereby boosting anomaly detection performance. Examples include TranAD [34] and DTADD [35]. Generative models, such as autoencoder [20], [33], [36] and generative adversarial network (GAN) [37], [38], [39], also demonstrate their strong ability in representation learning and reconstruction. However, these models often rely on a training set without anomalies to learn normal representation [37], which can reduce the general applicability of these methods.
- 2) Distance-based methods, such as local outlier factors (LOFs) [40] and isolation forests [41], which identify anomalies by high pairwise distances between series features. However, these methods struggle to handle complex, high-dimensional data due to the curse of dimensionality. As the dimensionality increases, the pairwise distance between any two points becomes nearly equal, making it difficult to distinguish between normal and anomalous points. Although some methods, such as DeepSVDD [42] and DeepSAD [43], extend the traditional distance-based approaches by incorporating deep learning methods, they still face challenges in handling extremely high-dimensional spaces and need a relative clean training set to learn a hypersphere to separate anomalies from normal data.
- 3) Density-based methods, which identify anomalies as observations with low density by density estimation. These approaches do not impose restrictions on data distribution and offer high adaptability. Classical density-based methods such as kernel density estimate [10] and GMMs [44] struggle with complex, high-dimensional data. DAGMM [14] improves the GMMs by combining it with autoencoder but still has limitations when applied to multivariate time series. Normalizing flows can address these issues [17], leveraging reversible and differentiable transformations to

map complex distributions into standard ones, thereby facilitating density estimation and anomaly detection [18], [19]. However, the performance of normalizing flow on time series detection has not been fully exploited. Further efforts should be made by exploring more temporal features of series and improving model robustness.

III. PRELIMINARY

In this section, we present our experiments and findings on the consistency feature of anomalies within decomposed series and provide some background information about normalizing flow.

A. Consistency Among Decomposed Time Series

We measured the density consistency among time series and their decomposed components. The density was estimated using masked autoregressive flows (MAFs) [15], a variant of normalizing flow. Specifically, we first progressively decomposed the series in each detection window (set to 60 timestep) into trend and seasonal components using average pooling [7]. The original series and their decomposed components were input into MAF to map distributions. We used the trained model to estimate the density of both the original and decomposed series. We then calculated the Kullback–Leibler (KL) divergence between their density estimation, with higher KL divergence indicating lower consistency. Experiments were conducted on five datasets: PSM [8], SWaT [45], WADI [46], MSL [31], and SMD [32].

We applied the bootstrap method [47] to calculate 95% confidence intervals (CIs) of the KL divergence sum of normal and abnormal series. The number of bootstrap samples was set to align with the number of specific series. We recorded the CIs, means, and medians of the KL divergence sums in Table I. Additionally, Fig. 3 provides a comparative visualization for normal and abnormal series. The scatterplot compares detailed KL divergence between the different components. The boxplot compares the KL divergence sum between normal and abnormal series.

Overall, the results show that the density consistency for normal time series is significantly higher than that for anomalous series. As shown in Table I and Fig. 3, normal time series generally exhibit lower KL divergence with

TABLE I
STATISTIC OF KL DIVERGENCE SUM AMONG DECOMPOSED TIME SERIES

Metric Dataset	Normal				Abnormal			
	Number	Mean	Median	Confidence interval	Number	Mean	Median	Confidence interval
PSM [8]	2172	0.226	0.121	(0.208, 0.245)	1336	5.017	0.662	(3.248, 7.134)
SWaT [45]	8469	0.067	0.003	(0.051, 0.087)	524	34.050	0.191	(27.069, 42.317)
SMD [32]	896	0.094	0.031	(0.071, 0.130)	238	3.689	0.532	(1.672, 5.845)
MSL [31]	2730	0.146	0.081	(0.081, 0.272)	214	291.569	0.086	(12.381, 5454.851)
WADI [46]	6551	0.001	0.001	(0.001, 0.002)	356	73.179	0.003	(29.115, 122.732)

Note: We record the statistic of KL divergence sum between normal and abnormal decomposed time series, including number of series, mean, median and 95% confidence interval of KL divergence sum. Note that the SMD dataset contains 38 sub-datasets, and we randomly selected one sub-dataset (machine1-1) for density estimation and consistency measurement.

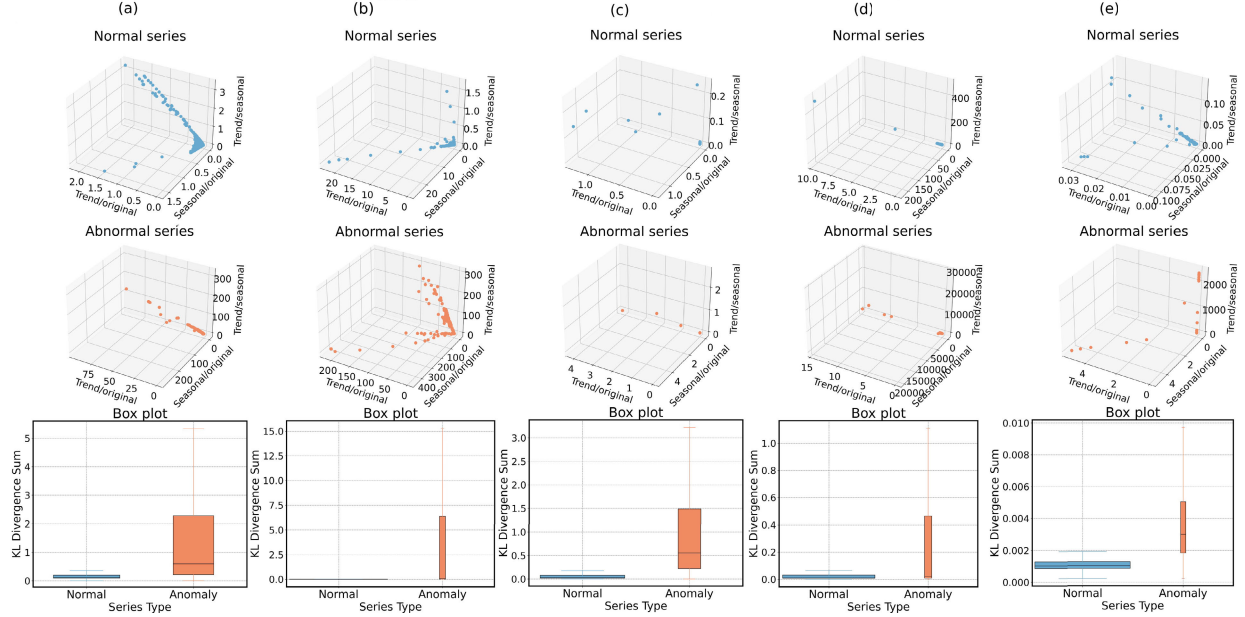


Fig. 3. Comparison of KL divergence among normal and abnormal decomposed series on PSM, SWaT, SMD, MSL, and WADI datasets. The 3-D scatterplot shows the KL divergence of density estimation between trend/original, seasonal/original, and seasonal/trend series. Boxplot compares the KL divergence sum for normal (blue) and abnormal (orange) series, with extreme outliers filtered out and box width proportional to the number of series. (a) PSM. (b) SWaT. (c) SMD machine-1-1. (d) MSL. (e) WADI.

narrower CIs, indicating higher density consistency. For example, in the PSM dataset, the mean of KL divergence sum for normal series is 0.226, with 95% CI of 0.208–0.245, while the mean for anomalous series shows a significant increase as 5.017, with 95% CI as 3.248–7.134. Similar trends are observed in the SWaT, SMD, MSL, and WADI datasets. In some datasets, the mean of KL divergence for anomalous series can be significantly affected by extreme outliers, resulting in a high mean value while the median remains much lower. For instance, the median for anomalous series in MSL is 0.086, similar to that of normal series (0.081). This occurs because anomaly labels are applied to entire windows of multivariate series instead of individual series. As shown in Fig. 4(a), several series (such as Features 2, 3, 4, 5, and 7) are labeled as anomalous but display the same pattern and exhibit low KL divergence.

We also find that density consistency has limitations in helping discriminate anomalies within the datasets where normal series exhibit inconsistent patterns. For instance, in Fig. 4(b), Feature 6 is a normal series but displays inconsistent spike-like patterns, resulting in high KL divergence among the

data density distribution of its decomposed components. In these cases, the decomposed components of a normal series may also show low-density consistency due to their inherent instability, which reduces the effectiveness of using density consistency as a distinguishing factor of an anomaly. Despite this limitation, density consistency can differentiate frequency variations of the spike-like signals, measuring 13.82 for the higher-frequency spike [Feature 6 in Fig. 4(a)] and 8.18 for the lower-frequency spike [Feature 6 in Fig. 4(b)]. This also indicates the effectiveness of density consistency in detecting frequency-related anomalies within spike-like signals.

To conclude, although using density consistency among decomposed series to represent anomalies has certain limitations, it remains a valuable and generalizable representation to identify anomalies.

B. Normalizing Flow Models

Density-based methods identify anomalies as data points with low density in the data distribution. Normalizing flow

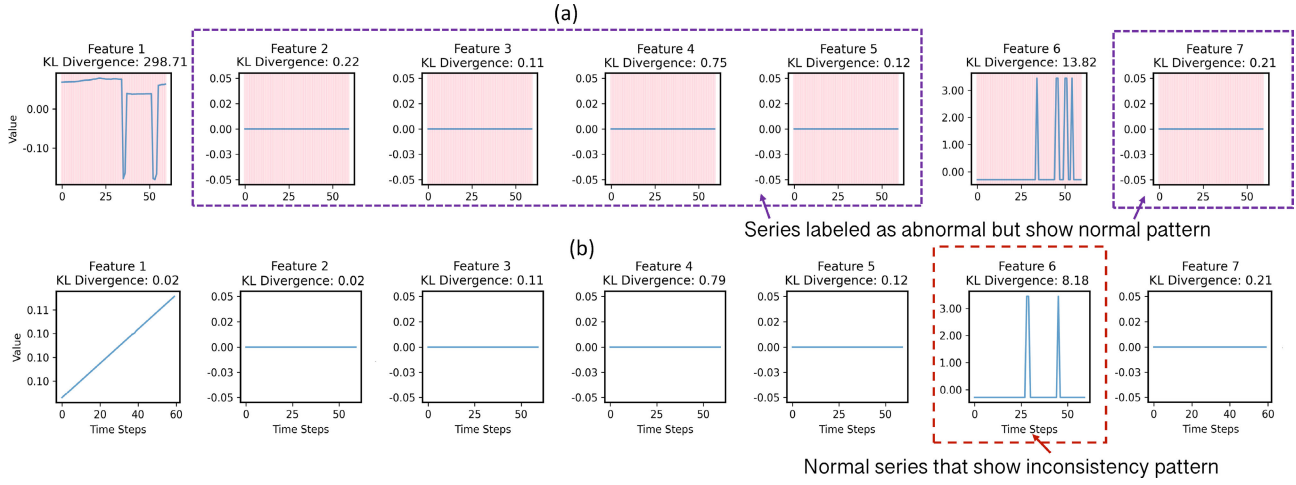


Fig. 4. Illustration of (a) abnormal and (b) normal series selected from MSL dataset, with KL divergence sum among their decomposed series marked above figures. In MSL, the time series are named by features, labeled as Feature 1, Feature 2, Feature 3, Feature 4, and so on, corresponding to different sensor measurements.

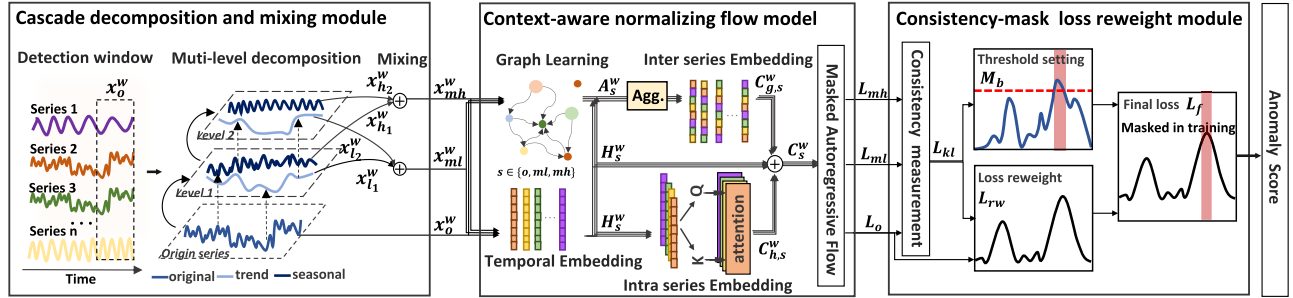


Fig. 5. Overview of ConFlow. Time series in a detection window x_o^w are first decomposed and mixed to extract seasonal x_h^w and trend components x_{ml}^w . These components are then processed to encode conditional information C_s^w between and within series, followed by MAF that is used for mapping distributions. $s \in \{o, ml, mh\}$ represents the original, trend, and seasonal components, and Agg. denote aggregate information from different series. After that, density consistency among decomposed series is measured and used for loss reweight and training mask, where the reweight loss represents anomaly scores.

models facilitate this process by mapping data from complex distribution $x \in \mathbb{R}^D$ into simpler standard distribution $z \in \mathbb{R}^D$, resulting in more straightforward density estimation [48] and easier anomaly detection. By normalizing flow, we can transform the observed data and compute its likelihood using the change of variable formula

$$P_X(x) = P_Z(z) \left| \det \frac{\partial f_\theta}{\partial x^T} \right| \quad (1)$$

where $P_Z(z)$ is the target distribution, $z = f_\theta(x)$ is the transformation function, and $|\det(\partial f_\theta / \partial x^T)|$ is the Jacobian determinant of the transformation. Anomalies can be detected when the likelihood $P_X(x)$ falls below a threshold ζ

$$P_X(x) = P_Z(z) \left| \det \frac{\partial f_\theta}{\partial x^T} \right| < \zeta \quad (2)$$

where ζ is chosen based on the expected density of normal data.

IV. PROPOSED MODEL

A. Problem Statement

We focus on unsupervised multivariate time series anomaly detection. The dataset consists of multiple unlabeled time series $X = [X_1, X_2, \dots, X_K]$, where K denotes the number

of series. No restrictions are placed on the training data. Detection is performed on a detection window of size T , sliding with a step size of S . If any point within the sliding window w is anomalous, the window is marked as anomalous. Our goal is to learn the conditional normalizing flow $F : (X|C) \rightarrow Z$ (C is the conditional information) to estimate the density of time series. The estimated density is used for anomaly detection.

B. Overall Structure

Fig. 5 illustrates the proposed model, comprising three primary components.

- 1) *Cascade decomposition and mixing module*: This module aims to extract refined trend and seasonal time series components from the original time series $x_o^w \in \mathbb{R}^{K \times T}$. It first decomposes x_o^w into trend and seasonal components through a multilevel decomposition. Then, the decomposed trend and seasonal time series at different decomposition levels are combined to represent each component.
- 2) *Context-aware normalizing flow model*: This module aims to encode information and mapping distributions for time series. First, the temporal embedding of series is

encoded through a long short-term memory (LSTM) network. The series relationship is then quantified by their similarity, constructing a graph that is used for aggregating inter-series information. Next, a self-attention mechanism is applied to the temporal embedding to extract intra-series contextual information. Finally, this information is integrated as the conditional input for MAF, used for mapping Gaussian distributions Z_o , Z_{ml} , and Z_{mh} .

- 3) *Consistency-mask loss reweight module*: This module aims to utilize the density inconsistency of anomalies to enhance model robustness and discriminative ability. Since anomalous time series shows lower density consistency among its decomposed components, consistency is directly used for loss reweight to enhance representation of anomalies. Additionally, series with extremely high inconsistency scores are masked during training to enhance model tolerance against anomalies.

The detailed descriptions of these modules are provided as follows.

C. Time Series Cascade Decomposition and Mixing

Inspired by Autoformer [7], in each detection window w , we decompose the series x_o^w into trend and seasonal components using average pooling. This process separates the low-frequency trend and high-frequency seasonality, formulated as

$$x_l^w = \text{Avg Pool}(\text{Padding}(x_o^w)) \quad (3)$$

$$x_h^w = x_o^w - x_l^w. \quad (4)$$

Here, x_l^w and x_h^w represent the trend and seasonal components, $\text{Avg Pool}(\cdot)$ denotes average pooling, $\text{Padding}(\cdot)$ represents reflection padding, which handles boundary conditions by mirroring edge values.

We perform a cascade decomposition to distill more nuanced information. Specifically, low-frequency series are decomposed to yield smoother patterns, while high-frequency series are further decomposed to extract more fluctuating patterns

$$x_{l_i}^w = \text{Avg Pool}(\text{Padding}(x_{l_{i-1}}^w)) \quad (5)$$

$$x_{h_i}^w = x_{h_{i-1}}^w - \text{Avg Pool}(\text{Padding}(x_{h_{i-1}}^w)) \quad (6)$$

where $x_{l_i}^w$ and $x_{h_i}^w$ represent the trend and seasonal components achieved at the i th decomposition level. The decomposed components across different levels are finally combined to represent the specific components

$$x_{ml}^w = \sum_{i=1}^c \alpha x_{l_i}^w \quad (7)$$

$$x_{mh}^w = \sum_{i=1}^c \alpha x_{h_i}^w \quad (8)$$

where $\alpha = 1/c$ represents the weighting factor for each level, and c is the number of decomposition levels.

D. Context-Aware Normalizing Flow

After decomposition, time series and their decomposed components are processed through context-aware normalizing flow, which first captures information between and within time series. The embedded information is then input into MAF to map distributions.

For each detection window w , LSTM is first applied on series to obtain a temporal encoding

$$H_{s,t}^w = \text{LSTM}(x_{s,t}^w, H_{s,t-1}^w). \quad (9)$$

$H_{s,t}^w \in \mathbb{R}^{K \times D}$ represents the temporal encoding at t , with $s \in \{o, ml, mh\}$ representing the original, trend, and seasonal time series, $t \in \{1, 2, \dots, T\}$ representing the time step, and D representing the hidden dimension.

To quantify the relationships among multivariate series, we follow MTGFLOW [49] and model series as nodes and similarities between nodes as directed edges, which composes a directed graph allowing for self-loops. For each pair of series $x_{s,i}^w, x_{s,j}^w \in \mathbb{R}^T$ (where $i, j \in \{1, 2, \dots, K\}$), the similarity directed from node i to node j is given by

$$e_{s,ij}^w = \frac{\exp\left((x_{s,i}^w M_{s,i})(x_{s,j}^w M_{s,j})^\top / \sqrt{T}\right)}{\sum_{k=1}^K \exp\left((x_{s,i}^w M_{s,i})(x_{s,k}^w M_{s,k})^\top / \sqrt{T}\right)} \quad (10)$$

where $M_{s,i}, M_{s,j}, M_{s,k} \in \mathbb{R}^{T \times T}$ are learnable matrices for linear projection, and T is for numerical stability. The function $\exp(\cdot)$ transforms the similarity scores into positive values, facilitating normalization. The similarity values $e_{s,ij}^w$ form the adjacency matrix of the graph

$$A_s^w = \begin{bmatrix} e_{s,11}^w & e_{s,12}^w & \cdots & e_{s,1K}^w \\ e_{s,21}^w & e_{s,22}^w & \cdots & e_{s,2K}^w \\ \vdots & \vdots & \ddots & \vdots \\ e_{s,K1}^w & e_{s,K2}^w & \cdots & e_{s,KK}^w \end{bmatrix}. \quad (11)$$

We use the adjacency matrix to aggregate temporal information from high-related series, defined as

$$C_{g,s,t}^w = A_s^w H_{s,t}^w. \quad (12)$$

Considering anomalies typically exhibit different correlation with their adjacent timesteps compared with the normal ones, a self-attention mechanism is applied on the window-level temporal embedding $H_s^w \in \mathbb{R}^{K \times T \times D}$ to capture the intra-series contextual information within each series. For series k , its contextual information can be formulated as

$$C_{h,s,k}^w = \text{softmax}\left(\frac{(H_{s,k}^w Q_s^w)(H_{s,k}^w K_s^w)^\top}{\sqrt{D}}\right) H_{s,k}^w V_s^w \quad (13)$$

where $Q_s^w, K_s^w, V_s^w \in \mathbb{R}^{D \times D}$ are the learnable matrices of linear transformation, and D is the model dimensionality and used for numerical stability.

The final conditional information at time t is obtained by integrating the above information

$$C_{s,t}^w = \text{ReLU}(C_{g,s,t}^w W_{s,1} + H_{s,t-1}^w W_{s,2} + C_{h,s,t}^w W_{s,3}) W_{s,4} \quad (14)$$

where $C_{s,t}^w \in \mathbb{R}^{K \times T \times D}$, and $W_{s,1}, W_{s,2}, W_{s,3}, W_{s,4} \in \mathbb{R}^{D \times D}$ represent the weights for inter-series aggregation,

temporal information, intra-series information, and sum of the embedding, respectively.

Based on the conditional encoding, we stack MAF models to map distribution for each series. The density estimation for series k in detection window w is given by

$$P_{X_{s,k}}(x_{s,k}^w) = \prod_{t=1}^T P_{Z_{s,k,t}}(f_{\theta}^{s,k}(x_{s,k,t}^w | C_{s,k,t})) \cdot J_{s,k,t} \quad (15)$$

$$J_{s,k,t} = \left| \det \left(\frac{\partial f_{\theta}^{s,k}}{\partial (x_{s,k,t}^w)^{\top}} \right) \right|. \quad (16)$$

Here, $f_{\theta}^{s,k}$ represents the autoregressive function of MAF, $J_{s,k,t}$ is the Jacobian determinant of $f_{\theta}^{s,k}$, and $C_{s,k,t}$ represents the conditional information for series k at t . $Z_{s,k}$ is the target mapping distribution that follows a normal distribution:

$$Z_{s,k} \sim N(\mu_{s,k}, I), \quad \mu_{s,k} \sim N(0, 1) \quad (17)$$

with mean $\mu_{s,k}$ sampled from a standard normal distribution, and I representing the identity covariance matrix.

E. Consistency-Mask Loss Reweight

The training objective of flow models is to maximize the joint probability of observed data, aligning the generative distribution closer to the true distribution. Therefore, the basic loss function in each mini-batch can be formulated as

$$\min_{\vartheta} \mathcal{L}_s(\vartheta) = \frac{1}{NK} \sum_{w=1}^N \sum_{k=1}^K (-\log(P_{X_{s,k}}(x_{s,k}^w))) \quad (18)$$

where $s \in \{o, ml, mh\}$, ϑ represents all model parameters, N is the batch size, and K is the number of multivariate series.

For each detection window, we get three density estimations (original, trend, and seasonal). Density consistency for $x_{o,k}^w$ is measured by summing the KL divergences between the density estimations of original, trend, and seasonal components

$$\mathcal{L}_{kl}(x_{o,k}^w) = \sum_{\substack{i,j \in \{o, ml, mh\} \\ i \neq j}} \text{KL}(P_{X_{i,k}}(x_{i,k}^w) \| P_{X_{j,k}}(x_{j,k}^w)). \quad (19)$$

Since anomalous decomposed series often exhibit inconsistent density estimation, consistency is directly used for loss reweight, penalizing inconsistency and enhancing the anomaly score. The reweight loss function can be denoted as

$$\min_{\vartheta} \mathcal{L}_{rw}(\vartheta) = \frac{1}{NK} \sum_{w=1}^N \sum_{k=1}^K (-\log(P_{X_{o,k}}(x_{o,k}^w)) + \mathcal{L}_{kl}(x_{o,k}^w)). \quad (20)$$

Even though flow models can be built with contaminated training data, we found that clean training data can still improve the performance of the model with one-class classification [49]. To mitigate the impact of extremely anomalous samples during training, we propose a mask strategy based on the density inconsistency of anomalies. In each mini-batch, series with high consistency loss $\mathcal{L}_{kl}(x_{o,k}^w)$ are masked during training to reduce their impact. The masking threshold β can be set as a fixed value or percentile value. Specifically,

Algorithm 1 Training Process of ConFlow

Input: Training set $\{X\}_{i=1}^N$, window size T , stride size S , decomposition level c , learning rate η , mask threshold β , and *epoch_num*;

Output: Optimized parameters θ and anomaly scores for each detection window;

Initialize: Parameters θ ; detection windows (based on T and S);

for $e \in [1, \text{epoch_num}]$ **do**

- 1) Use cascade decomposition and mixing module to generate time series set $\{x_o^w, x_{ml}^w, x_{mh}^w\}$ via Eqs.(3) to (8);
- 2) Encode inter- and intra-series information using the context-aware encoder for time series via Eqs.(9) to (14);
- 3) Generate the density estimations P_{X_o} , $P_{X_{ml}}$, and $P_{X_{mh}}$ for time series using MAF via Eqs.(15) to (17);
- 4) Measure the consistency $\mathcal{L}_{kl}$ among the density estimations based on Eq.(19);
- 5) Reweight the loss $\mathcal{L}_o$ with $\mathcal{L}_{kl}$ based on Eq.(20). The reweight loss serves as the anomaly score;
- 6) Mask series with $\mathcal{L}_{kl}$ exceeds the threshold β , and update the model parameters θ ;

a Boolean tensor M_b is generated to mask series with a consistency score exceeding β during training.

Therefore, in each mini-batch, the final loss can be formulated as

$$\min_{\vartheta} \mathcal{L}_f(\vartheta) = \frac{1}{NK} \sum_{w=1}^N \sum_{k=1}^K M_b(-\log(P_{X_{o,k}}(x_{o,k}^w)) + \mathcal{L}_{kl}(x_{o,k}^w)). \quad (21)$$

F. Training and Testing

The training process of ConFlow can be summarized in Algorithm 1.

The anomaly score of each series at time t is measured by its reweight density

$$S_{x_{o,k}^w} = -\log(P_{X_{o,k}}(x_{o,k,t}^w)) + \mathcal{L}_{kl}(x_{o,k,t}^w). \quad (22)$$

The anomaly score for each detection window is the average of the series involved

$$S_w = \frac{1}{K} \sum_{k=1}^K (-\log(P_{X_{o,k}}(x_{o,k}^w)) + \mathcal{L}_{kl}(x_{o,k}^w)). \quad (23)$$

Anomalies can be identified by setting a threshold ζ . The series with scores exceeding ζ are classified as anomalies. ζ is determined based on the overall scores of the training series, for instance, by using the interquartile range (IQR) method, setting ζ as $Q_3 + 1.5 \times (Q_3 - Q_1)$. Q_3 and Q_1 represent the 75th and 25th percentiles of the overall score, respectively.

G. Complexity Analysis

We measure the asymptotic upper bound of computational complexity for each detection window using big O notation [50], represented by a function of the input and omitting

constant factors and lower order terms. For each detection window, the input of the model is related to the number of series K , the window size T , and the model dimension D .

We first analyze the time complexity of ConFlow, denoted as $T(K, T, D)$. The cascade decomposition and mixing module has a time complexity linear with the input time series, represented as $O(K \times T)$. The context-aware encoding module encompasses several operations having certain time complexity: temporal encoding by LSTM $O(K \times T \times D^2)$ [51], graph construction via similarity calculation $O(K^2 \times T \times D)$, inter-series information aggregation $O(K^2 \times T \times D)$, and intra-series correlation using self-attention $O(K \times T^2 \times D)$ [52]. Subsequently, the MAF maps the distributions with a time complexity of $O(K \times T \times D^2)$ [53]. The consistency-mask loss reweight module involves: mask generation $O(K \times T \times \log(KT))$ and loss reweight operations $O(K \times T)$. The dominant terms in the time complexity depend on the relative sizes of K , T , and D , and can be denoted as: $T(K, T, D) = O(\max(K \times T^2 \times D, K^2 \times T \times D, K \times T \times D^2))$.

Space complexity measures the storage usage as the input increases. The cascade decomposition and mixing module requires $O(K \times T)$ space. The context-aware encoding module involves: temporal encoding by LSTM $O(K \times T \times D)$ [51], graph construction $O(K^2 \times T \times D)$, inter-series information aggregation $O(K \times T \times D)$, and intra-series correlation calculation $O(K \times T^2 \times D)$. The MAF module utilizes $O(K \times T \times D)$ space. Both the mask generation and the loss reweight module require $O(K \times T)$ space. The dominant terms in the space complexity also depend on the relative sizes of K , T , and D , which can be denoted as: $S(K, T, D) = O(\max(K \times T^2 \times D, K^2 \times T \times D))$.

V. EXPERIMENTS

This section presents various experiments that answer the following questions about our model ConFlow.

- Q1: Accuracy:* How accurately does our method detect anomalies compared to other baselines? Can our model pinpoint anomalous portions of the time series?
- Q2: Robustness:* How robust is our model when dealing with contaminated data with different ratios? How do different hyperparameters impact model performance?
- Q3: Explainability:* How and why do the modules designed in our model improve anomaly discriminative ability?

A. Experimental Details

We begin by detailing our experimental setup, covering dataset configurations, baseline models, and related settings in experiments.

1) *Dataset Setup:* We selected five publicly available datasets to evaluate model performance. SWaT [45] measures 51 sensors from a real-world industrial water treatment facility. PSM [8] comprises 25 distinct features of server nodes from an eBay application. MSL [31] consists of data from 55 sensors on NASA's Mars rover. SMD [32] contains 38 small-scale

TABLE II
DATASET INFORMATION

Dataset	Series	Training set		Testing set	
		Timestamps	Ratio	Timestamps	Ratio
SWaT	51	269951	17.7	89984	5.2
PSM	25	52704	23.1	35137	34.6
MSL	55	44237	14.7	29492	4.3
SMD	38	425052	4.2	283368	4.1
WADI	123	103680	6.4	69121	4.6

Note: Series and ratio represents the number of sensors and anomaly ratio of dataset, respectively.

datasets collected from 38 machines within an internet company. WADI [46] includes data from 123 actuators in a water distribution testbed.

Instead of using the conventional clean training set in most previous work [22], we followed the dataset partitioning strategy proposed by GANF [18] to obtain a contaminated but realistic training set. Specifically, we divided the original test set into training, validation, and testing sets by a ratio of 6:2:2. In this way, all datasets contain both normal and abnormal sequences, better reflecting real-world scenarios. Detailed statistics on the training and testing datasets, including timestamps and anomaly ratios, are provided in Table II.

2) *Baselines:* To assess the performance of ConFlow, we compared it against eight SOTA methods commonly used in multivariate time series anomaly detection. These baselines were selected to represent three core categories of anomaly detection techniques mentioned in Section II: distance-based, reconstruction-based, and density-based methods. Specifically, DeepSVDD and DeepSAD are distance-based methods that map normal time series into a hypersphere, identifying the data outside the hypersphere as anomalies. ALOCC, USAD, and FITS are reconstruction-based methods that trained on clean data to learn normal representations, detecting anomalies by reconstruction errors. Lastly, DAGMM, GANF, and MTGFLOW are density-based methods that align closely with ConFlow by identifying the data points distributed in low-density regions as anomalies. These methods were chosen for their representativeness and evaluated under a unified dataset setting.

A brief introduction to each baseline is as follows.

- 1) *DeepSVDD* [42]: A distance-based method that maps normal data into a predefined hypersphere using a deep neural network, with data outside this boundary detected as anomalies.
- 2) *DeepSAD* [43]: An extension of DeepSVDD utilizes labeled data to enhance anomaly detection performance.
- 3) *ALOCC* [54]: A reconstruction-based approach uses GANs to reconstruct time series so that anomalies are identified based on reconstruction error.
- 4) *USAD* [55]: A reconstruction-based method employs a two-phase autoencoder architecture, detecting anomalies based on reconstruction deviations.
- 5) *FITS* [56]: A reconstructed-based methods that empowered by representation ability of frequency interpolation.
- 6) *DAGMM* [14]: A density-based method that combines a deep autoencoder with a GMM to enhance the density estimation ability.

TABLE III
EXPERIMENTAL PARAMETERS OF CONFLOW IN COMPARATIVE EXPERIMENTS

Parameters Dataset	Window size	Window stride	Weight Decay	Learning rate	Hidden size	Block size	Batch size	Mask ratio	Decomposition level
SWaT	60	10	5×10^{-4}	2×10^{-3}	32	1	512	$Q_3 + 1.5 \times IQR$	3
PSM						2	256		2
MSL						2	256		2
SMD						2	256		2
WADI						2	256		3

Note: Block size represents the number of blocks stacked in the MAF model, hidden size refers to the hidden layer size for each block in MAF.

7) *GANF* [18]: A density-based model that integrates Bayesian Networks with flow models, identifying data in low-density regions as anomalies.

8) *MTGFLOW* [19]: An improved variant of GANF, mapping different time series into different distributions to improve detection accuracy.

3) *Experimental Setting:* All experiments were conducted on an Ubuntu 20.04.5 LTS server with 192 GB of memory, an Intel Xeon Gold 6242R CPU at 3.10 GHz, and a GeForce RTX 3090 graphics card. In the comparison study, we set the anomaly detection window size to 60 and the stride size to 10 for all experiments. Our model was implemented in PyTorch version 1.7.1 with CUDA 12.4. Each model was trained for 40 epochs using the Adam optimizer with a learning rate of 0.002 and a weight decay of 0.0005. The hidden dimension for all modules was set to 32. For the batch size, we used 512 for the SWaT dataset and 256 for the other datasets. Similarly, the decomposition level was set to 3 for both SWaT and WADI datasets, and to 2 for the others. Detailed parameter settings of our model are provided in Table III. For the baseline methods, we used publicly available implementations under the contaminated training setting. The code for DeepSVDD, DeepSAD, ALOCC, and GANF was obtained from [57], while the code for USAD, DAGMM, FITS, and MTGFLOW was obtained from [58], [59], [60], and [61], respectively. We retained their original network parameters and trained them for 40 epochs to ensure a fair comparison.

For the parameter selection of our model, we aligned the general parameters such as window size, stride size, learning rate, weight decay, batch size, block size, and hidden dimension with those of the SOTA density-based method MTGFLOW (as recorded in Table III). After fixing these parameters, we treated others, such as decomposition level and mask ratio, as tunable hyperparameters, whose values are optimized through extensive experiments. Specifically, we evaluated decomposition levels from the set $\{1, 2, 3, 4\}$ on all datasets, while the mask ratio was chosen from $\{Q_2, Q_3, Q_3 + 1.5 \times IQR\}$ for each mini-batch. The optimal decomposition levels identified were 3 for SWaT and WADI, and 2 for PSM, MSL, and SMD. The best mask ratio was consistently found to be $Q_3 + 1.5 \times IQR$ for all datasets. Besides, in our parameter study, we show how block size, window size, decomposition level, and mask ratio affect performance of our model.

4) *Evaluation Metrics:* We assessed model performance using the area under the receiver operating characteristic curve

(AUC-ROC), precision (P), recall (R), and *F1* score (*F1*), evaluating different aspects of model performance. AUC-ROC captures the trade-off between true positive and false positive rates across different thresholds, with higher values indicating better discriminative performance. AUC-ROC is evaluated based on the anomaly scores assigned by the model. Following the methodology of MTGFLOW [19], we reported the average and standard deviation of AUC-ROC over five random seed settings.

In contrast, precision (P), recall (R), and *F1* score (*F1*) evaluate model detection performance based on a specific point-level threshold. A detection window is classified as abnormal if any timestep within the window exceeds the threshold. Consistent with MTGFLOW [19], we set the threshold as $Q_3 + 1.5 \times (Q_3 - Q_1)$ of the anomaly scores from the training sets. Point-adjusted strategy was employed [20], [56] when evaluating P, R, and *F1*.

B. Performance Comparison

1) *Performance Analysis:* We compared ConFlow with eight baselines, and the results are recorded in Table IV (AUC-ROC) and Table V (P, R, and *F1* scores). ConFlow consistently outperformed all baselines in AUC-ROC across all datasets, achieving improvements over the second-best method (MTGFLOW) of 5.5% on SWaT, 2.6% on PSM, 1.1% on MSL, 1.0% on SMD, and 0.7% on WADI. ConFlow also demonstrated stable performance across five random experiments, with the standard deviation of AUC-ROC remaining at a low level. In terms of *F1* score (Table V), ConFlow achieved competitive performance against the best-performing baselines (FITS and MTGFLOW), with improvements of 6.03% on SWaT, 0.75% on MSL, 0.32% on SMD, and 1.14% on WADI. ConFlow attained the best recall across all datasets by leveraging the inconsistency feature of anomalies, without sacrificing much precision.

We further analyzed the performance of the models according to their categories. First, distance-based methods such as DeepSVDD and DeepSAD showed poor performance on all metrics across datasets because they lacked mechanisms to encode temporal dependencies. Moreover, DeepSVDD requires clean datasets to learn a hypersphere that surrounds normal representations. However, the contaminated training sets in our setting disrupted the construction of a discriminative hypersphere. Second, reconstruction-based methods like ALOCC and USAD also exhibited poor performance as they lacked the ability to capture subtle variations among time series. Conversely, FITS outperformed ALOCC and

TABLE IV
COMPARISON OF AUC-ROC (%) BETWEEN CONFLOW AND BASELINE MODELS

Dataset \ Model	Distance-based		Reconstruction-based			Density-based			
	DeepSVDD	DeepSAD	ALOCC	USAD	FITS	DAGMM	GANF	MTGFLOW	ConFlow
SWaT	66.8±2.0	75.5±2.4	77.5±2.3	78.5±1.0	80.1±1.1	72.8±3.0	78.5±0.7	84.8±1.5	90.3±0.6
PSM	67.5±1.4	73.2±3.3	71.8±1.3	78.5±1.2	81.5±1.6	64.5±2.6	81.5±1.5	85.7±1.5	88.3±0.8
MSL	60.8±0.4	61.4±0.6	60.5±0.9	57.0±0.1	60.4±0.7	56.5±2.6	64.5±1.9	<u>67.2±1.7</u>	68.3±1.3
SMD	75.5±15.5	85.9±11.1	80.5±11.1	86.5±11.1	88.1±7.2	78.5±9.2	89.5±7.8	<u>91.3±7.6</u>	92.3±6.7
WADI	83.5±1.6	83.8±1.8	83.3±1.8	86.1±0.9	89.0±0.4	77.2±0.9	90.3±1.0	<u>91.9±1.1</u>	92.6±0.3

Note: The AUC-ROC(%) is presented as mean $\pm$ standard deviation. A higher mean represents a better ability to discriminate between normal and abnormal series, and a lower deviation represents more stable performance. The best and second-best results are marked in **bold** and with underline, respectively.

TABLE V
COMPARISON OF PRECISION (%), RECALL (%), AND $F1$ SCORES (%) BETWEEN CONFLOW AND MTGFLOW

Model \ Dataset	SWaT			PSM			MSL			SMD			WADI		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
DeepSVDD	55.63	68.56	61.42	58.08	69.21	63.13	40.42	61.81	48.83	40.08	78.63	53.07	58.65	86.53	69.86
DeepSAD	68.75	77.39	72.80	69.29	77.30	73.07	43.33	56.79	49.19	51.68	87.05	64.73	63.83	82.97	72.17
ALOCC	70.34	77.84	73.82	68.24	75.02	71.50	41.37	60.51	49.14	67.81	82.14	74.27	76.34	81.95	78.86
USAD	70.85	79.43	74.94	70.51	81.02	75.47	41.57	63.39	50.23	72.10	88.17	79.33	78.24	85.10	81.59
FITS	93.21	68.86	79.13	99.59	87.26	93.02	48.21	59.82	53.35	90.91	<u>92.65</u>	<u>91.81</u>	94.57	82.08	87.85
DAGMM	69.95	73.73	71.76	61.21	63.92	62.54	45.29	57.43	50.62	71.08	77.26	74.03	70.91	76.65	73.71
GANF	80.75	75.21	77.88	83.45	71.13	76.80	46.26	59.82	52.17	89.78	87.58	88.66	91.25	89.25	90.29
MTGFLOW	86.13	<u>82.25</u>	<u>84.23</u>	<u>92.10</u>	82.87	87.24	58.96	<u>63.79</u>	<u>61.28</u>	91.88	90.23	91.07	<u>92.97</u>	<u>91.08</u>	<u>92.05</u>
ConFlow	<u>89.06</u>	91.38	90.26	90.85	94.52	<u>92.65</u>	<u>52.88</u>	75.00	62.03	<u>91.18</u>	93.15	92.13	92.63	93.88	93.19

Note: P denotes precision, R denotes recall, and F1 denotes F1 score. Metrics for SMD are reported as mean value of the sub-datasets. The best and second-best results are marked in **bold** and with underline, respectively.

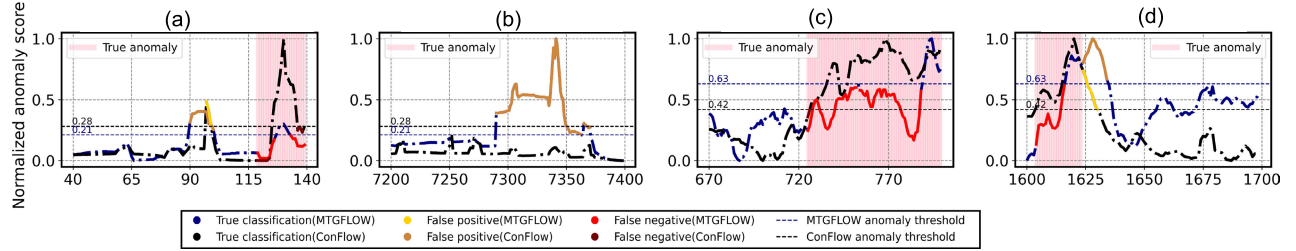


Fig. 6. Comparison of anomaly scores rated by MTGFLOW and ConFlow for four selected series segments. True classification consists of the segments of true positive and true negative. Anomaly thresholds for ConFlow (blue) and MTGFLOW (black) are marked in each sub-figure. (a) SWaT Segment 1. (b) SWaT Segment 2. (c) PSM Segment 1. (d) PSM Segment 2.

USAD by leveraging the robust representation capabilities of frequency interpolation, achieving the highest $F1$ score on the PSM dataset. However, its discriminative ability on different thresholds (AUC-ROC) remained unsatisfactory, primarily because a contaminated training set compromised its discriminative performance. Third, density-based methods, with the exception of DAGMM, showed good discriminative performance (AUC-ROC). The unsatisfactory performance of DAGMM was mainly due to its shortcomings in capturing relationships between time series. GANF and MTGFLOW exhibited threshold-invariant discriminative ability since they assign anomaly scores based on the probability of the time series. However, these methods had limitations in exploring the generalizable and contextual features of anomalies, degrading the detection ability. ConFlow outperformed GANF and MTGFLOW on both $F1$ scores and AUC-ROC across all five datasets, attributed to its utilization of cascade decomposition consistency and contextual information embeddings. Furthermore, ConFlow demonstrated consistent performance across

five different random seeds (standard deviation of AUC-ROC decreases 0.9% on SMD compared with MTGFLOW), primarily due to the consistency-mask strategy that enhance model robustness.

We further visualized the normalized anomaly scores rated by ConFlow and MTGFLOW in selected time segments, as illustrated in Fig. 6. In each sub-figure, the true anomalous segments are highlighted with a red background. The blue and black dashed lines represent the anomaly scores from MTGFLOW and ConFlow, with horizontal dashed lines denoting the anomaly thresholds. False positives and false negatives are marked in orange and red for MTGFLOW, and yellow and brown for ConFlow. As shown in Fig. 6(a), ConFlow was more sensitive to anomalies with more true positives and true negatives, which is due to the contextual embedding that models intra-series dependency. MTGFLOW had a higher false positive rate and a false negative rate [as shown in Fig. 6(b) and (c)] due to the lack of representation of anomalies.

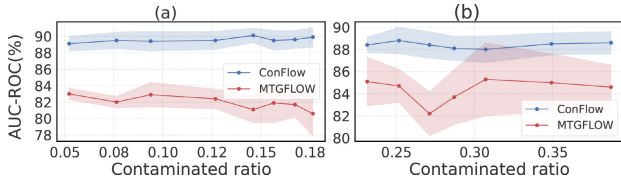


Fig. 7. Performance comparison of AUC-ROC between MTGFLOW and ConFlow under different training contamination ratios. Line represents the mean value and shadow represents the standard deviation. (a) SWaT. (b) PSM.

2) *Robustness Comparison Under Different Contaminated Ratios*: We divided the training data of SWaT and PSM into training sets with varying contamination ratios to evaluate the model's robustness under different contamination rates, while keeping the testing set fixed. The training set anomaly ratio for SWaT ranges from 5.2% to 17.6%, and for PSM, it ranges from 23.2% to 38.7%. The anomaly ratios of the testing sets are 5.2% for SWaT and 30.2% for PSM.

We compared the model performance of ConFlow with MTGFLOW under training set with different contamination ratios. The results are shown in Fig. 7. The shaded areas in the figures indicate the standard deviations of model performance across five random seeds. Generally, ConFlow demonstrates more robust and superior performance. Compared to MTGFLOW, our model consistently maintains higher AUC-ROC scores. In the SWaT dataset [Fig. 7(a)], as the contamination ratio increases, the standard deviation of MTGFLOW shows an increasing trend, while that of ConFlow remains low. In PSM dataset [Fig. 7(b)], the performance of ConFlow is more stable compared to MTGFLOW, with the standard deviation consistently remaining at a low level.

C. Result Visualization

Anomalies can be traced back to individual sensors based on anomaly scores. We selected six sensors in SWaT and visualized their anomaly scores and corresponding original time series (Fig. 8). The thresholds are marked on the right side of each subplot. Notably, some sensors exhibited similar responses to the same anomaly. For example, AIT502 and FIT504 showed related analogous reactions. This similarity may suggest that these sensors were highly connected to each other and experienced cascade failures. Additionally, anomalies do not occur across all sensors. For example, LTT101 shows an anomalous pattern on the first and fourth true labels, while others do not. LIT301 shows an anomalous pattern on the last true label, whereas other sensors remain unaffected.

D. Module Ablation and Visualization

In this section, we perform an ablation study to evaluate the effectiveness of the module design.

1) *Module Ablation*: We evaluated the effectiveness of the modules designed in ConFlow by testing various combinations on the PSM and SWaT datasets. The modules include context-aware encoding (C), loss reweight (L), and consistency-mask (M). ConFlow without specific modules is denoted as ConFlow/(module abbreviation). For example,

ConFlow/(C, L, M) indicates the model without all three modules.

The ablation results are presented in Table VI. The base model, ConFlow/(C, L, M), achieved performances of 84.7% on SWaT and 85.3% on PSM. Among the individual modules, incorporating only the loss reweight module [ConFlow/(C, M)] provided the most significant improvement, enhancing performance by 3.5% on SWaT and 1.7% on PSM. This indicates that the loss reweight module (L) is the most effective single module, especially on the SWaT dataset. Combining the loss reweight module with the context-aware encoding module [ConFlow/(M)] resulted in further improvements, achieving a 4.6% increase on SWaT and 2.1% on PSM. Including all modules (ConFlow) yielded the best performance, with improvements of 5.6% on SWaT and 3.0% on PSM compared to the base model. These results demonstrate that the loss reweight module significantly enhances performance, and combining it with other modules leads to further gains, particularly on the SWaT dataset.

2) *Visualization of Context-Aware Encoding Module*: We visualized the intra-series contextual encoding of four selected series (Series 1, Series 2, Series 3, and Series 24) from the PSM dataset, as shown in Fig. 9. For each detection window, the attention embeddings were calculated using the normalized sum of the attention scores. As depicted in the figure, different segments of the time series exhibited distinct patterns, and their contextual embeddings evolved over time. Notably, the context-aware encoding for anomalies displayed different patterns.

Specifically, for Series 1, both the time series values and the attention embeddings showed a relatively stable pattern most of the time. However, the attention scores significantly increased during certain time intervals (i.e., timestamps 13 000–14 000, 16 000–17 000, and 28 000–29 000), where a potential contextual and point anomalies appeared during these periods. Similarly, Series 2, 3, and 24 also exhibit unique attention embeddings when anomalies occur.

E. Parameter Study

We tested different hyperparameters to evaluate the robustness of the model.

1) *Decomposition Level, Window Size, and Flow Block*: We analyzed the impact of various hyperparameters, including decomposition level, window size, and the number of flow model blocks.

First, we fixed the window size at 60 and varied the decomposition levels and the number of flow blocks to assess their effects on model performance. The results are shown in Table VII. A larger model (with more flow blocks) may lead to overfitting and does not necessarily improve detection performance, especially when the number of blocks exceeds two. Moreover, a higher decomposition level does not always enhance model performance, and the optimal decomposition level varies across datasets. Specifically, with a fixed number of flow blocks, a two-level decomposition is optimal for the PSM dataset, while a three-level decomposition is best for the SWaT dataset.

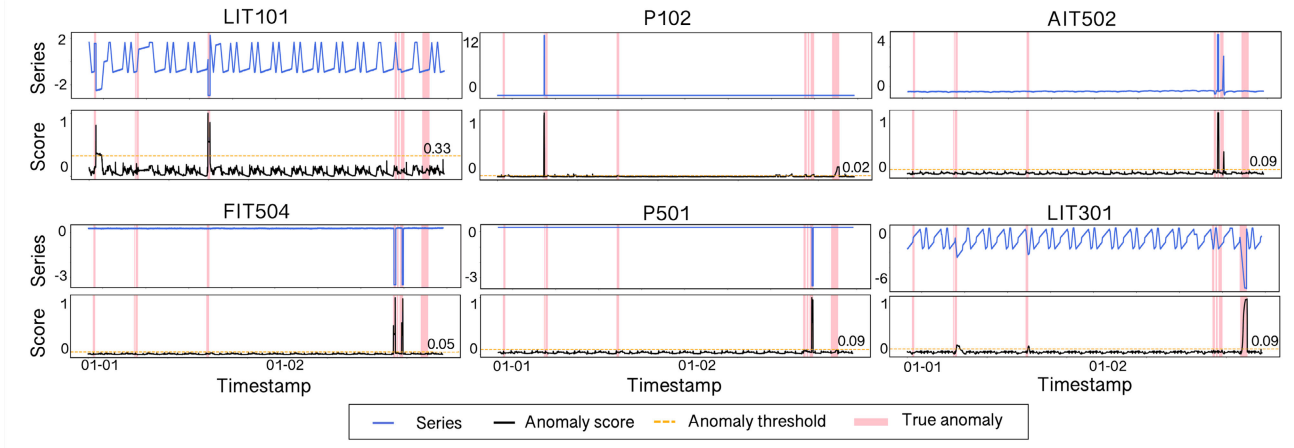


Fig. 8. Six series selected from SWaT and their corresponding anomaly scores.

TABLE VI
STUDY OF MODULE ABLATION

Model	Module			Dataset	
	C	L	M	SWaT	PSM
ConFlow/(C, L, M)	✗	✗	✗	84.7±2.0	85.3±2.7
ConFlow/(L, M)	✓	✗	✗	86.0±1.4	86.2±1.5
ConFlow/(C, L)	✗	✗	✓	85.2±1.3	85.9±1.1
ConFlow/(C, M)	✗	✓	✗	88.2±1.7	87.0±1.0
ConFlow/(M)	✓	✓	✗	89.3±0.9	87.4±0.8
ConFlow/(L)	✓	✗	✓	85.4±1.1	86.4±1.5
ConFlow/(C)	✗	✓	✓	89.1±1.3	87.1±1.0
ConFlow	✓	✓	✓	90.3±0.6	88.3±0.8

Note: C, L, and M represent the context-aware encoding, loss reweight, and consistency-mask modules, respectively.

TABLE VII

PARAMETER STUDY OF BLOCK NUMBER AND DECOMPOSITION LEVEL

DL	Blocks	1	2	3	4
SWaT	1	89.7±1.1	89.0±0.9	88.2±1.5	86.0±2.1
	2	89.9±1.2	89.7±1.1	87.4±2.4	85.2±2.1
	3	90.3±0.6	90.4±0.7	87.5±2.6	86.1±2.7
	4	89.3±1.1	89.7±0.9	87.5±2.5	85.1±2.9
PSM	1	83.7±1.4	86.4±1.2	86.4±1.3	87.6±0.3
	2	86.9±0.7	88.3±0.8	88.0±1.2	87.8±0.5
	3	85.0±1.7	86.9±1.1	87.2±0.9	86.6±1.4
	4	82.9±2.1	85.8±1.8	84.9±2.3	86.6±2.1

Note: DL represents the decomposition level, blocks represents the number of flow block.

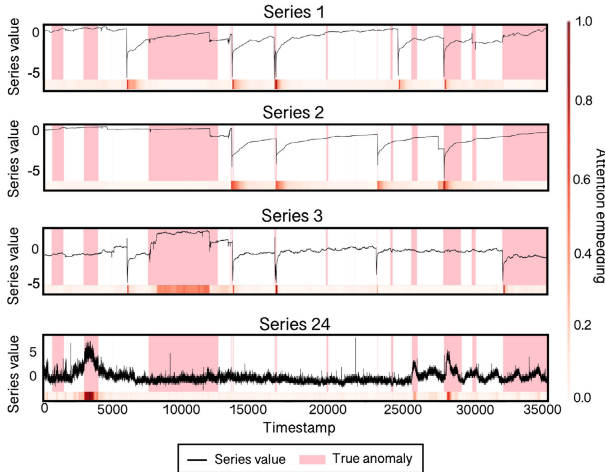


Fig. 9. Four series selected from PSM test set and corresponding attention embedding.

Next, we fixed the decomposition level (3 for SWaT and 2 for PSM) and conducted experiments with various detection window sizes and number of flow blocks to assess model robustness. The results are presented in Table VIII. For the PSM dataset, detection performance degrades as the window size increases, whereas for the SWaT dataset, performance improves with larger window sizes. Specifically, PSM achieves

TABLE VIII

PARAMETER STUDY OF BLOCK NUMBER AND WINDOW SIZE

Window	Blocks	1	2	3	4
SWaT	40	89.5±0.7	89.6±0.9	87.2±1.1	85.7±2.8
	60	90.3±0.6	90.4±0.7	87.5±2.6	86.1±2.7
	80	90.4±0.9	90.4±1.1	88.2±2.7	85.9±3.1
	100	89.8±1.0	90.6±0.5	87.3±1.3	85.1±2.5
	120	89.2±0.6	90.2±0.7	87.2±1.4	85.6±3.1
PSM	40	87.2±1.6	88.4±1.3	88.7±0.5	88.8±0.7
	60	86.9±0.7	88.3±0.8	88.0±1.2	87.8±0.5
	80	86.6±1.0	87.3±0.3	86.1±1.2	87.4±0.8
	100	85.5±0.8	85.8±1.2	85.8±1.2	86.0±0.5
	120	84.9±1.2	85.1±0.7	85.3±1.1	85.5±1.4

Note: Window represents the window size.

the best performance with four flow blocks and a window size of 40, while SWaT demonstrates optimal performance with two flow blocks and a window size of 100. These findings suggest that selecting an appropriate window size is crucial and should be tailored to the characteristics of each specific dataset.

2) *Consistency-Mask Ratio Analysis:* We further evaluated the model's performance using different consistency mask ratios (Table IX). Q_2 represents the 50th percentile of the consistency score in each mini-batch, while Q_3 represents the

TABLE IX
PARAMETER STUDY OF MASK RATIO

Dataset \ MR	Q_2	Q_3	$Q_3 + 1.5 \times IQR$
SWaT	89.5 ± 0.6	90.0 ± 0.7	90.3 ± 0.6
PSM	87.6 ± 1.5	88.1 ± 1.2	88.3 ± 0.8

Note: MR represents the mask ratio in each mini-batch. Q_2 represents the median, Q_3 represents the third quartile, and IQR represents the interquartile range ($1.5 \times (Q_3 - Q_1)$).

75th percentile. For both datasets, increasing the mask ratio (in other words, lowering the mask threshold) does not improve model performance. The model achieves the best performance when the mask threshold is set at $Q_3 + 1.5 \times (Q_3 - Q_1)$. This phenomenon may be attributed to the contamination ratio of the original training set. The training set contamination ratio is 23.1% for PSM and 17.7% for SWaT. A high mask ratio may reduce the proportion of normal series in training, leading to performance degradation.

VI. CONCLUSION

In this work, we proposed ConFlow, an unsupervised anomaly detection approach for multivariate time series. ConFlow is a density-based approach that utilizes the density inconsistency and contextual features of anomalies to enhance the discriminative ability of the Flow model. Extensive experiments demonstrate that our method outperforms other baselines in detecting anomalies in multivariate time series. ConFlow can accurately pinpoint anomalous positions within time series and maintains stable performance across varying contamination ratios.

While ConFlow demonstrates good performance across most datasets, we identified a limitation when applied ConFlow to datasets where normal time series exhibit inconsistent patterns (such as spike-like signals in MSL). Future work will focus on enhancing the model's robustness to such natural inconsistencies. Additionally, we are exploring methods for detecting early anomalies which are similar with normal behavior.

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